

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
CODE: CSA5107

COURSE

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

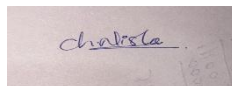
APPLICATION- BASED QUESTIONS

Code of Conduct:

I Chodam Nisha certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature appears to be 'Chodam Nisha'.

APPLICATION- BASED QUESTIONS

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Given Ciphertext:

“GSRH RH Z HVXIVG”

The intercepted message appears to be encrypted using a **simple substitution mechanism**. After examining the structure of the ciphertext, the **Atbash cipher** is the most likely technique used.

a) Deduce the most likely original message

The **Atbash cipher** is a classical substitution cipher in which the letters of the alphabet are reversed.

The basic mapping is:

Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	
-----------	---	---	---	---	---	---	---	---	---	---	---	---	---	--

Ciphertext	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	
------------	---	---	---	---	---	---	---	---	---	---	---	---	---	--

For the remaining letters:

Plaintext	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
-----------	---	---	---	---	---	---	---	---	---	---	---	---	---	--

Ciphertext	\M	L	K	J	I	H	G	F	E	D	C	B	A	
------------	----	---	---	---	---	---	---	---	---	---	---	---	---	--

Now decrypt the given message:

GSRH

- $G \rightarrow T$
- $S \rightarrow H$
- $R \rightarrow I$
- $H \rightarrow S$

Therefore:

GSRH → THIS

Similarly:

RH → IS

Z → A

HVXIVG → SECRET

Therefore, the complete original message is:

“THIS IS A SECRET.”

b) Justification

The message can be identified as an **Atbash cipher** by observing its letter patterns and applying the substitution rule.

1. Fixed substitution

In a substitution cipher, one letter is consistently replaced by another letter.

For Atbash:

A → Z, B → Y, C → X, ... G → T, H → S

For example:

GSRH → THIS

The same mapping is used throughout the message.

2. Word structure remains unchanged

The encrypted message contains:

GSRH | RH | Z | HVXIVG

The word lengths are:

4 | 2 | 1 | 6

After decryption:

THIS | IS | A | SECRET

The same word-length structure is maintained. This is a useful indication of a substitution cipher.

3. Single-letter word

The ciphertext contains the single-letter word “Z”.

In English, common one-letter words are **A** and **I**. Using the Atbash mapping:

Z → **A**

This supports the identification of the cipher.

4. Consistent result

When the Atbash rule is applied to every letter, the result forms a meaningful English sentence:

“THIS IS A SECRET.”

Therefore, the message is most likely encrypted using the **Atbash substitution cipher**.

Conclusion

The Atbash cipher is simple because it uses a fixed alphabet-reversal rule. Although it can hide a message from casual observation, it does not provide strong security against modern cryptanalysis.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario

During a digital forensic audit, investigators suspect that a **company logo** contains hidden confidential information. The image appears normal during casual inspection.

This situation can involve **digital image steganography**, where secret information is hidden inside an ordinary-looking image. CO1_ASSESSMENT 1

a) How can investigators detect hidden data?

Investigators should follow a systematic forensic process rather than immediately modifying or extracting information from the original file.

1. Examine the image file

First, investigators collect basic information about the suspicious image, such as:

- File format
- File size
- Image dimensions
- Creation and modification information

- Available metadata

An unusual file size or unexpected metadata may indicate that the image needs further examination.

However, these differences alone do not prove that hidden data exists.

2. Compare with the original image

If the original company logo is available, investigators can compare both versions.

They can compare:

- File size
- Image dimensions
- Metadata
- Pixel information
- Image structure

If the suspicious image contains unexplained changes, investigators can investigate those changes further.

3. Examine metadata

Image files may contain metadata that provides information about the file.

Investigators can check for:

- Author information
- Software used
- Dates
- Comments
- Other embedded information

Unexpected or suspicious metadata can provide useful forensic clues.

4. Perform pixel-level analysis

One common method of image steganography is **LSB (Least Significant Bit) steganography**.

Digital images are represented using numerical pixel values. Small changes can be made to the least significant bits to store hidden information.

For example:

Original:10110100

Modified: 10110101

Only the last bit has changed. Such a small change may not be easily noticed when viewing the image normally.

Therefore, investigators can examine the pixel values and look for unusual patterns.

5. Perform statistical analysis

Investigators can also examine the statistical properties of the image.

They may analyze:

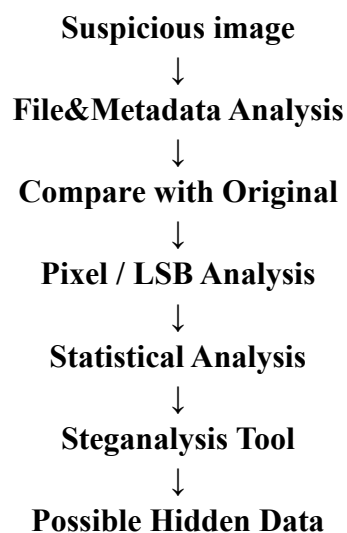
- Pixel distribution
- Color values
- LSB patterns
- Image noise
- Other unusual statistical patterns

If the suspicious image has patterns that are significantly different from a normal image, it may indicate possible steganographic modification.

6. Use steganalysis tools

Specialized forensic tools can be used to analyze suspicious images and detect possible hidden information.

A general investigation process is:



This approach helps investigators determine whether the image has been modified to carry hidden information.

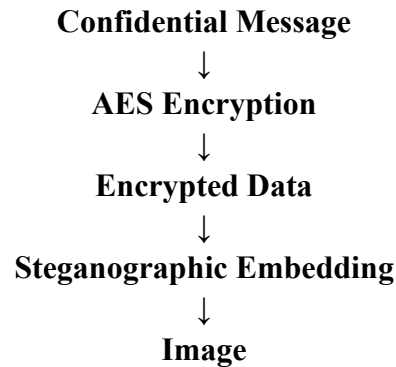
b) Two techniques to strengthen the security of hidden communications

1. Encrypt the information before hiding it

The first technique is to **encrypt the confidential information before embedding it inside the image.**

For example, AES can be used to encrypt the original message.

The process is:



Suppose the original information is:

“CONFIDENTIAL REPORT”

Instead of directly hiding this readable text, it is first converted into encrypted data.

If an attacker detects the hidden information and extracts it, the attacker will still have to decrypt it using the correct key.

Advantage

This provides two levels of protection:

1. **Steganography hides the existence of the information.**
2. **Encryption protects the actual information.**

Therefore, combining both techniques provides stronger confidentiality.

2. Use Adaptive Steganography

The second technique is **adaptive steganography**.

In a simple steganography method, data may be inserted into pixels using a predictable pattern. Such predictable changes can sometimes be detected through statistical analysis.

Adaptive steganography instead considers the characteristics of the image before selecting locations for hiding data.

For example:

- Smooth areas can be changed less.
- Complex or textured areas can be selected for embedding.
- Areas containing natural variations can be used to reduce noticeable changes.

This helps reduce obvious visual and statistical changes in the image.

Advantage

Adaptive embedding can make basic steganalysis more difficult because the hidden data is not inserted using a simple uniform pattern.

Final Conclusion

In this scenario, the company logo may be used as a cover image for confidential information. Investigators can detect possible hidden data by checking the file properties, metadata, original image comparison, pixel values, LSB patterns, statistical characteristics, and steganalysis tools.

For stronger protection, the confidential information can first be encrypted and then hidden inside the image. An adaptive steganography technique can also be used to make simple detection more difficult.

Thus, encryption protects the content, while steganography hides the existence of the communication. Combining both provides a stronger approach for protecting confidential information.